

WHEN THE SAME OBJECT IS NOT THE SAME: INTERFACE-CONDITIONED IDENTITY FOR ROBOT MA- NIPULATION

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Paper under double-blind review

ABSTRACT

Robot manipulation papers usually treat object identity as a property of the object, then ask a gripper, sensor, or active policy to estimate that fixed label. This abstraction becomes brittle when the gripper is part of the observation process: a pinch gripper may reveal stiffness and friction, a suction cup may reveal porosity and seal quality, and an enveloping hand may reveal size and compliance while collapsing distinctions that matter elsewhere. We argue that manipulation perception should represent object identity as an interface-conditioned partition of latent object states. For a gripper-action interface, two states have the same identity when the interface cannot distinguish them and no task-relevant controller needs to. We formalize this as an Interface-Conditioned Identity Partition (ICIP), prove a finite partition impossibility result showing when no single gripper-agnostic identity can be both observable and sufficient, and provide a small reproducible manipulation-perception experiment. In the experiment, global object IDs are sufficient but not observable from a single gripper channel; common gripper-agnostic labels are observable but not sufficient for control; ICIP recovers the observable and task-sufficient distinctions. The contribution is not a larger model or a new benchmark, but a change in the central representational variable for contact-rich manipulation.

1 INTRODUCTION

Object recognition is often written as if the world provides object labels and the robot merely estimates them. That picture is useful for static vision, but contact-rich manipulation makes the robot interface part of the measurement apparatus. A parallel jaw gripper, a suction cup, and a soft enveloping hand do not just execute different grasps; they create different observables. They collapse and separate different latent object states. If the label space is held fixed while the observation channel changes, a learner may be asked to infer distinctions that are physically unavailable to its gripper, or a planner may consume labels that omit distinctions its gripper needs.

This paper asks a narrow question: when should an object identity label be indexed by the gripper-action interface that produces the observations? The answer matters for tactile recognition, active perception, affordance learning, object-centric world models, and robot foundation models that hope to move representations across hands and tools. The dominant assumption in much prior work is that action helps reveal a fixed object identity (Bajcsy, 1988; Bohg et al., 2017; Pajarinen & Kyrki, 2019), or that touch is another modality for estimating object or property labels (Calandra et al., 2018; Yuan et al., 2017; Chitta et al., 2010). Those views are not wrong. The missing case is the one where changing the gripper changes the identity relation that is observable and useful.

We propose Interface-Conditioned Identity Partitions (ICIP). For an interface g —a gripper plus a family of probing or manipulation actions—ICIP is the quotient of latent object states that are observationally indistinguishable and task-equivalent under g . This is a deliberately modest claim. We do not deny that a complete latent physical state may exist. We claim that a deployed robot often cannot and need not identify that state. The useful identity abstraction is the coarsest one that still supports the manipulation decisions available to the current interface.

Our contributions are:

1. A formal statement of gripper-dependent identity as a partition-lattice problem over latent object states.
2. A finite impossibility result: when two interfaces induce non-identical minimal observable-sufficient partitions, no single identity partition can be minimal for both; under stronger non-nesting, any universal compromise is either unobservable or insufficient for at least one interface.
3. A reproducible synthetic manipulation-perception environment with pinch, suction, and enveloping grippers that separates three abstractions: global object ID, common gripper-agnostic label, and ICIP.
4. A broad literature audit over 1000 Crossref records, with 300 serious-skim entries, 225 deep-read metadata/abstract entries, and 100 hostile prior-work entries. The audit is released with this submission and explicitly marks its metadata-level limitations.

2 RELATED WORK

Interactive and active perception. Active perception has long treated action as a way to improve state estimation (Bajcsy, 1988). Interactive perception in robotics makes the coupling between action and perception explicit, especially when manipulation reveals shape, pose, articulation, or hidden structure (Katz et al., 2011; Bohg et al., 2017; Hausman et al., 2015; Pajarinen & Kyrki, 2019). This line is closest to our work because it rejects passive sensing. However, it usually preserves a fixed target variable: the robot chooses actions to reveal an object, pose, property, or model. ICIP instead asks whether the target identity relation itself should change once the action/gripper channel is specified.

Tactile and visuotactile object recognition. Tactile sensing has been used for object recognition, material inference, grasp correction, pose estimation, and regrasping (Chitta et al., 2010; Calandra et al., 2018; Yuan et al., 2017; Lee et al., 2019). Recent visuotactile and in-hand systems increasingly treat contact as a central signal rather than an auxiliary cue. The hostile prior-work set includes recent papers on slip-aware in-hand manipulation, shared visuotactile pose estimation, and tactile object recognition. These works make tactile recognition less novel; they do not by themselves define the label space as a quotient induced by a gripper-action interface.

Grasping, grippers, and affordances. Data-driven grasp synthesis and grasp-quality prediction have produced strong object-to-action mappings (Bohg et al., 2014; Mahler et al., 2017; Pinto & Gupta, 2016). Gripper design papers make the morphology of the hand central to what can be grasped, sensed, or controlled (RUTH Gripper Authors, 2021; Slip-Aware Manipulation Authors, 2025; Soft Gripper Authors, 2026). Affordance learning binds objects and actions (Gibson, 1979; Do et al., 2018). ICIP is adjacent but different: affordances describe what actions are possible, while ICIP describes which object distinctions are observable and sufficient for a given interface.

Object-centric robot representations. Object-centric world models and manipulation representations seek reusable variables for planning and policy learning (Greff et al., 2020; Veerapaneni et al., 2020). The pressure in that literature often favors invariance: learn object variables that transfer across views, tasks, and embodiments. ICIP marks a boundary condition for that ambition. Invariance across grippers can be harmful when the relevant partitions are non-nested: one interface collapses distinctions another must preserve.

3 INTERFACE-CONDITIONED IDENTITY

Let S be a finite set of latent object states. An interface g is a gripper together with a family of actions and sensors. It induces an observation kernel $O_g(y \mid s)$ and a task equivalence relation over states. We write partitions with the refinement order $\mathcal{A} \preceq \mathcal{B}$ when every block of $\mathcal{A}$ is contained in a block of $\mathcal{B}$. Thus $\mathcal{A}$ is finer than $\mathcal{B}$.

Definition 1 (Observation partition). *The observation partition $\mathcal{O}_g$ groups states $s, s' \in S$ when $O_g(\cdot \mid s) = O_g(\cdot \mid s')$. No identity label observable through g can split a block of $\mathcal{O}_g$.*

Definition 2 (Task partition). *The task partition $\mathcal{T}_g$ groups states that require the same manipulation decision under interface g . An identity partition is sufficient for g only if it does not merge states from different blocks of $\mathcal{T}_g$.*

Table 1: Synthetic evidence. Global IDs are task-sufficient but not observable from a single gripper channel; common gripper-agnostic labels are observable but lose control distinctions; ICIP is both observable and sufficient in this constructed world. Global-ID action success is high because the controller only uses the quotient-relevant part of the predicted label, which is the point of quotienting.

Gripper	Identity target	Classes	Label acc.	Action succ.	Global-ID ceiling
Enveloping	Global ID	96	11.7%	100.0%	12.5%
Enveloping	Common label	6	100.0%	50.0%	12.5%
Enveloping	ICIP	12	100.0%	100.0%	12.5%
Pinch	Global ID	96	53.1%	100.0%	50.0%
Pinch	Common label	6	100.0%	37.5%	50.0%
Pinch	ICIP	48	100.0%	100.0%	50.0%
Suction	Global ID	96	25.2%	100.0%	25.0%
Suction	Common label	6	100.0%	75.0%	25.0%
Suction	ICIP	24	100.0%	100.0%	25.0%

An identity partition $\mathcal{I}$ is observable for g if $\mathcal{O}_g \preceq \mathcal{I}$. It is sufficient for g if $\mathcal{I} \preceq \mathcal{T}_g$. Hence an observable and sufficient identity for g must satisfy

$$\mathcal{O}_g \preceq \mathcal{I} \preceq \mathcal{T}_g. \quad (1)$$

When $\mathcal{O}_g \preceq \mathcal{T}_g$, a natural minimal interface-conditioned identity is $\mathcal{I}_g = \mathcal{O}_g$ if the observation partition is already task-sufficient, or the coarsest observable partition that still refines $\mathcal{T}_g$ when redundant observations exist.

Definition 3 (ICIP). *An Interface-Conditioned Identity Partition for interface g is a partition $\mathcal{I}_g$ that is observable through g , sufficient for g 's task partition, and minimal among partitions satisfying both constraints.*

Proposition 1 (No universal minimal identity under distinct ICIPs). *Let g and h be two interfaces with unique minimal observable-sufficient partitions $\mathcal{I}_g$ and $\mathcal{I}_h$. If $\mathcal{I}_g \neq \mathcal{I}_h$, no single partition $\mathcal{I}$ is the unique minimal observable-sufficient identity for both interfaces. If, additionally, $\mathcal{I}_g$ and $\mathcal{I}_h$ are non-comparable under refinement and each equals its interface's required task partition, then any single compromise partition is either unobservable or insufficient for at least one of the two interfaces.*

Proof sketch. If $\mathcal{I}$ were the unique minimal observable-sufficient identity for both, uniqueness gives $\mathcal{I} = \mathcal{I}_g$ and $\mathcal{I} = \mathcal{I}_h$, contradicting $\mathcal{I}_g \neq \mathcal{I}_h$. For the stronger statement, consider any partition $\mathcal{I}$. To be observable and sufficient for g when $\mathcal{O}_g = \mathcal{T}_g = \mathcal{I}_g$, it must satisfy $\mathcal{I}_g \preceq \mathcal{I} \preceq \mathcal{I}_g$, hence $\mathcal{I} = \mathcal{I}_g$. The same argument for h requires $\mathcal{I} = \mathcal{I}_h$. Non-comparability rules out a shared equality, so at least one observability or sufficiency constraint fails. $\square$

4 SYNTHETIC MANIPULATION-PERCEPTION EVIDENCE

We instantiate the partition conflict in a finite toy world designed to be easy to audit. There are 96 latent objects with attributes size, shape, stiffness, porosity, friction, and mass. Three interfaces observe different subsets: pinch observes size, shape, stiffness, friction, and mass; suction observes size, shape, porosity, and mass; enveloping observes size, shape, and stiffness. Each gripper has an optimal controller that depends only on its observable task-relevant attributes.

We compare three identity targets. Global ID is the full six-attribute object label. Common label is the gripper-agnostic visual label containing only size and shape. ICIP is the gripper-specific quotient over the attributes observed and needed by that interface. For each gripper and identity target, a nearest-centroid classifier is trained on noisy observations. We then measure label accuracy, action success after mapping predicted labels to majority optimal controllers, and the Bayes upper bound for global ID under the gripper's observation partition.

Table 1 shows the intended separation. For pinch, suction, and enveloping grippers, global-ID accuracy is approximately bounded by the number of gripper-visible quotient classes divided by the

Table 2: Partition relations in the toy world. Pinch refines enveloping, but suction is non-nested with both pinch and enveloping. This is the structural condition that makes a single minimal identity inappropriate.

Pair	First refines second	Second refines first
Pinch vs. Suction	no	no
Enveloping vs. Pinch	no	yes
Enveloping vs. Suction	no	no

96 global states: 50.0%, 25.0%, and 12.5%. The common label is perfectly observable because all grippers see size and shape, but it is insufficient for control: action success is 37.5%, 75.0%, and 50.0%. ICIP has perfect label accuracy and action success because, by construction, it keeps exactly the distinctions each interface can observe and needs.

The experiment is not meant to prove real-world prevalence. It demonstrates a failure mode that a larger model, uncertainty wrapper, or verifier would not remove: the wrong label space can be unobservable or insufficient before estimation begins.

5 LITERATURE AUDIT AND NOVELTY BOUNDARY

We ran a 1000-paper landscape sweep over manipulation perception queries, then extracted a 300-paper serious skim, a 225-paper deeper metadata/abstract read, and a 100-paper hostile prior-work set. The full matrix records, for each entry, the claimed problem, mechanism, hidden assumptions, fixed variables, ignored failures, what it makes less novel, and what it leaves open. The audit was conducted under API limits: Crossref was available, while OpenAlex and Semantic Scholar returned HTTP 429. Therefore the audit is useful for breadth and hostility, but not a final full-PDF related-work substitute.

The hostile set made several weak moves unacceptable. A bigger tactile model would only estimate a fixed label better. Active probing would choose actions for a fixed identity variable. A multi-gripper benchmark would expose the issue but not supply the mechanism. Affordance learning comes closest, yet it can still leave object identity fixed and treat action as a property query. The remaining novelty boundary is the quotient mechanism: identity is the observable-and-sufficient partition induced by the interface.

6 LIMITATIONS

This paper is a formal and mechanistic argument with synthetic evidence. It has no real robot trials. The toy world is deliberately discrete and favorable to ICIP because the observation and task partitions are hand-specified. The literature sweep is broad but metadata-limited. The theorem does not say universal latent states are impossible; it says deployable identity labels that must be both observable and task-sufficient may need to be interface-indexed. For a full submission, the next necessary step is a real multi-gripper dataset where visually identical or globally labeled objects split and collapse differently under pinch, suction, and enveloping interactions.

7 CONCLUSION

Manipulation perception should not always ask a gripper to estimate a fixed object identity. When interaction changes observables, the gripper is part of the identity relation. ICIP gives a compact way to express this: object identity is a quotient over latent states induced by the observation and control needs of a gripper-action interface. The formal result and runnable evidence show why a single gripper-agnostic identity can be the wrong abstraction even before model capacity or data scale enter the discussion.

REPRODUCIBILITY STATEMENT

The repository contains the literature sweep scripts, generated matrix, synthesis documents, and the full synthetic experiment. Run `python experiments/run_icip_experiment.py` from the repository root to reproduce the CSV metrics and LaTeX table snippets. No external Python packages are required for the experiment.

REFERENCES

- Ruzena Bajcsy. Active perception. *Proceedings of the IEEE*, 76(8):966–1005, 1988.
- Jeannette Bohg, Antonio Morales, Tamim Asfour, and Danica Kragic. Data-driven grasp synthesis - a survey. *IEEE Transactions on Robotics*, 30(2):289–309, 2014.
- Jeannette Bohg, Karol Hausman, Bharath Sankaran, Oliver Brock, Danica Kragic, Stefan Schaal, and Gaurav S. Sukhatme. Interactive perception: Leveraging action in perception and perception in action. *IEEE Transactions on Robotics*, 33(6):1273–1291, 2017.
- Roberto Calandra, Andrew Owens, Dinesh Upadhyaya, Wenzhen Yuan, Justin Lin, Edward H. Adelson, and Sergey Levine. More than a feeling: Learning to grasp and regrasp using vision and touch. *IEEE Robotics and Automation Letters*, 3(4):3300–3307, 2018.
- Sachin Chitta, Jurgen Sturm, Mario Piccoli, and Wolfram Burgard. Tactile object class and internal state recognition for mobile manipulation. In *IEEE International Conference on Robotics and Automation*, pp. 2342–2348, 2010.
- Thanh-Toan Do, Anh Nguyen, and Ian Reid. Affordancenet: An end-to-end deep learning approach for object affordance detection. In *IEEE International Conference on Robotics and Automation*, pp. 5882–5889, 2018.
- James J. Gibson. *The Ecological Approach to Visual Perception*. Houghton Mifflin, 1979.
- Klaus Greff, Sjoerd van Steenkiste, and Jurgen Schmidhuber. Binding in neural networks: A survey. In *arXiv preprint arXiv:2012.05208*, 2020.
- Karol Hausman, Scott Niekum, Sarah Osentoski, and Gaurav S. Sukhatme. Active articulation model estimation through interactive perception. In *IEEE International Conference on Robotics and Automation*, pp. 3305–3312, 2015.
- Dov Katz, J. Andrew Kenney, and Oliver Brock. Interactive segmentation, tracking, and kinematic modeling of unknown 3d articulated objects. In *IEEE International Conference on Robotics and Automation*, pp. 5003–5010, 2011.
- Michelle A. Lee, Yuke Zhu, Krishnan Srinivasan, Parth Shah, Silvio Savarese, Li Fei-Fei, Animesh Garg, and Jeannette Bohg. Making sense of vision and touch: Self-supervised learning of multimodal representations for contact-rich tasks. In *IEEE International Conference on Robotics and Automation*, pp. 8943–8950, 2019.
- Jeffrey Mahler, Jacky Liang, Sherdil Niyaz, Michael Laskey, Richard Doan, Xinyu Liu, Juan Aparicio Ojea, and Ken Goldberg. Dex-net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics. In *Robotics: Science and Systems*, 2017.
- Joni Pajarinen and Ville Kyrki. Active object recognition and pose estimation using deep reinforcement learning. *IEEE Robotics and Automation Letters*, 4(2):1749–1756, 2019.
- Lerrel Pinto and Abhinav Gupta. Supersizing self-supervision: Learning to grasp from 50k tries and 700 robot hours. In *IEEE International Conference on Robotics and Automation*, pp. 3406–3413, 2016.
- RUTH Gripper Authors. Systematic object-invariant in-hand manipulation via reconfigurable under-actuation: Introducing the ruth gripper. *The International Journal of Robotics Research*, 2021. doi: 10.1177/02783649211048929.

- Slip-Aware Manipulation Authors. Perception, control, and hardware for in-hand slip-aware object manipulation with parallel grippers. *The International Journal of Robotics Research*, 2025. doi: 10.1177/02783649251397549.
- Soft Gripper Authors. Bioinspired swallowing soft gripper with toroidal optical waveguides for multimodal interactive perception. *Soft Robotics*, 2026. doi: 10.1177/21695172261451702.
- Rishi Veerapaneni, John D. Co-Reyes, Michael Chang, Michael Janner, Chelsea Finn, Jiajun Wu, Joshua B. Tenenbaum, and Sergey Levine. Entity abstraction in visual model-based reinforcement learning. In *Conference on Robot Learning*, 2020.
- Wenzhen Yuan, Siyuan Dong, and Edward H. Adelson. Gelsight: High-resolution robot tactile sensors for estimating geometry and force. *Sensors*, 17(12):2762, 2017.